

Homework 3 – Theory

1

A is $m \times d$ with $m \gg d$, and $X = AA^T$ is $m \times m$. The black box takes a symmetric matrix and returns its eigenvalues and an orthonormal set of eigenvectors.

The idea is to run the black box on $A^T A$, which is only $d \times d$, and then turn its eigenvectors into eigenvectors of X by multiplying by A .

First, a quick fact about A . Since X is symmetric with d nonzero eigenvalues, $\text{rank}(X) = d$, and

$$d = \text{rank}(AA^T) \leq \text{rank}(A) \leq d,$$

so $\text{rank}(A) = d$ and $A^T A$ is invertible. Also

$$v^T (A^T A) v = \|Av\|^2 \geq 0$$

for any v , so the eigenvalues of $A^T A$ are all ≥ 0 , and since it is invertible they are all > 0 . We need this because we divide by them.

The algorithm:

1. Form $Y = A^T A$.
2. Run the black box on Y to get $Y = V\Lambda V^T$, with $V = [v_1 \cdots v_d]$ orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$, all $\lambda_i > 0$.
3. For each i , set $u_i = \frac{Av_i}{\sqrt{\lambda_i}}$.
4. Return the pairs (λ_i, u_i) . The other $m - d$ eigenvalues of X are 0, and their eigenvectors are anything orthogonal to the u_i (Gram–Schmidt gives a basis if we need one).

Now check the u_i . They have unit length, since

$$\|Av_i\|^2 = v_i^T A^T A v_i = \lambda_i v_i^T v_i = \lambda_i.$$

They are eigenvectors of X , since

$$Xu_i = \frac{A(A^T A) v_i}{\sqrt{\lambda_i}} = \frac{A(\lambda_i v_i)}{\sqrt{\lambda_i}} = \lambda_i u_i.$$

And they are orthogonal to each other, since

$$u_i^T u_j = \frac{v_i^T A^T A v_j}{\sqrt{\lambda_i \lambda_j}} = \frac{\lambda_j v_i^T v_j}{\sqrt{\lambda_i \lambda_j}} = 0 \quad \text{for } i \neq j.$$

So we have d orthonormal eigenvectors of X with nonzero eigenvalues. X only has d nonzero eigenvalues, so these are all of them, and everything else is 0.

For the cost, forming $A^T A$ is $O(md^2)$, the black box on a $d \times d$ matrix is $O(d^3)$, and computing $Av_1, \dots, Av_d$ is $O(md^2)$. In total that is $O(md^2 + d^3)$, compared to $O(m^3)$ for running the black box on X directly, and we never form an $m \times m$ matrix.

2

(a)

True. Linear regression picks w to minimize

$$\frac{1}{m} \sum_{i=1}^m \left(w^\top x^i - y^i \right)^2,$$

which needs the labels y^i . It is fitting a function to labeled data, so it is supervised.

(b)

True. PCA picks orthonormal $v_1, \dots, v_k$ to maximize

$$\frac{1}{m} \sum_{j=1}^m \sum_{i=1}^k \langle x^j, v_i \rangle^2,$$

and only the inputs x^j appear. There are no labels, so it is unsupervised.

(c)

False, it is the other way around. SVD is a matrix factorization, $A = USV^\top$, and it works on any matrix. PCA is a procedure on a dataset: center the data, then take the top eigenvectors of the covariance matrix $\frac{1}{m} X^\top X$. PCA is usually computed with an SVD of the centered data matrix, since the right singular vectors of X are the eigenvectors of $X^\top X$, but that makes SVD a tool that PCA uses, not a special case of it.

(d)

Run PCA on the inputs first, then do the regression on the projected data.

Center the data, compute the SVD of the centered matrix $X = USV^\top$, keep the k columns V_k whose singular values are nonzero, and replace each input by

$$z = V_k^\top (x - \mu) \in \mathbb{R}^k.$$

Fit w by least squares on the pairs (z^i, y^i) , and project new inputs the same way before predicting.

Why this works: if two columns are linearly dependent, some direction in feature space has zero variance, so $X^\top X$ has a zero eigenvalue and the normal equations $X^\top X w = X^\top y$ have infinitely many solutions. In the height example the centimetre column is 2.54 times the inches column, so $v \propto (2.54, -1)$ gives $Xv = 0$. PCA drops that direction. The projected data is $Z = XV_k = U_k S_k$, and

$$Z^\top Z = S_k^2$$

is diagonal with positive entries, so it is invertible and

$$w = (Z^\top Z)^{-1} Z^\top y$$

is unique. The redundant feature is gone, so there is only one way to fit the data.